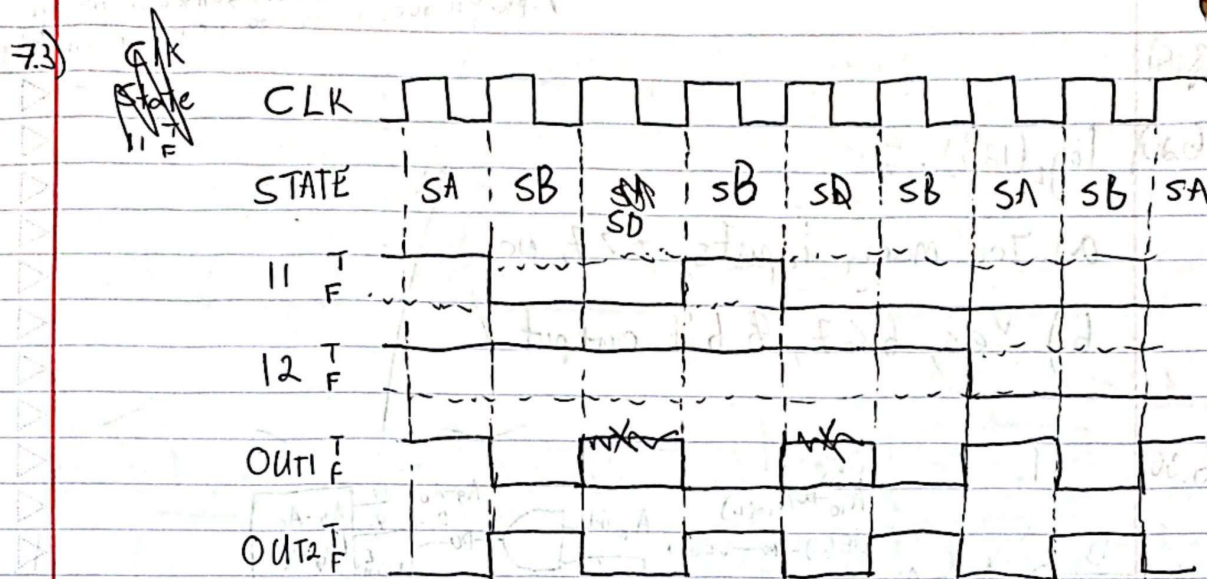

REQUIREMENTS NOT MET

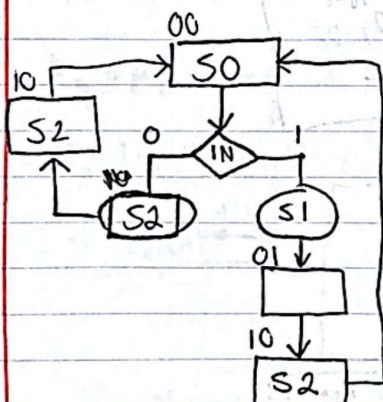
- Questions 7.15-7.23

PROBLEMS ENCOUNTERED

- N/A.

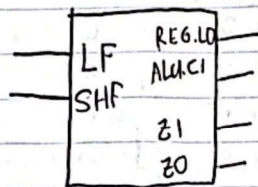


7.5) $S0 \rightarrow S1 \mid IN = 1$, $S0 \rightarrow S2 \mid IN = 0$
 $S1 \rightarrow S2$
 $S2 \rightarrow S0$

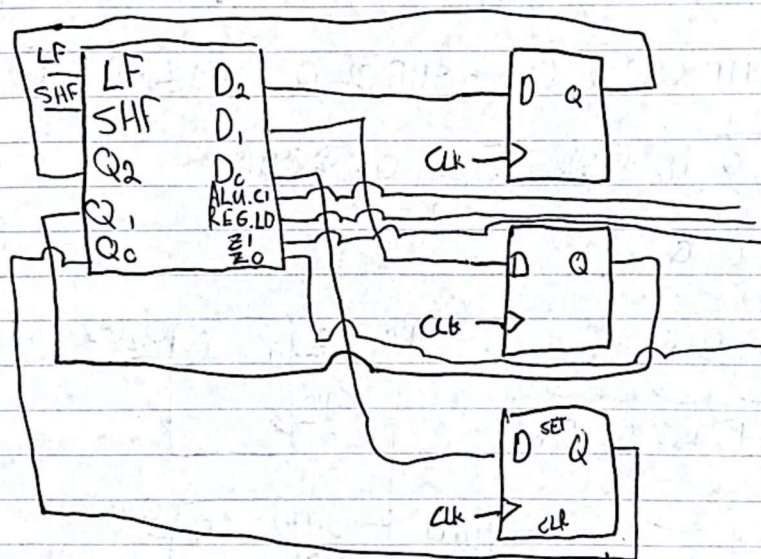


7.8) Inputs: LF, SHF

Outputs: REG.LD, ALU.CI, Z1, Z0



b: $S0 \rightarrow S4 = 5$ states, 3 ff



ALL ~~set~~/clr, tied to clr_L,
set to Vcc

7.9)

C:	LF	SHF	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	ALU.CI	REG.LD	ZI	ZO
0	0	0	0	0	0	0	0	1	0	1	0	0
0	0	0	0	0	1	0	1	0	1	0	0	0
0	0	0	0	1	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	0	0	1	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	1	0	0	1	0	1
0	1	0	1	1	1	0	1	1	1	0	1	0
1	0	0	0	0	0	0	1	1	0	1	0	0
+	+	+	+	+	+	+	+	+	+	+	+	+

$$DZ \leftarrow \overline{Q_2} \overline{Q_1} \overline{Q_0} \text{ SHF}$$

$$DI \leftarrow \overline{Q_2} \overline{Q_1} \overline{Q_0} + \overline{Q_2} \overline{Q_1} \overline{Q_0} \text{ LF SHF} + \text{SHF} \overline{Q_2} \overline{Q_1} \overline{Q_0} + \text{LF} \overline{Q_2} \overline{Q_1} \overline{Q_0}$$

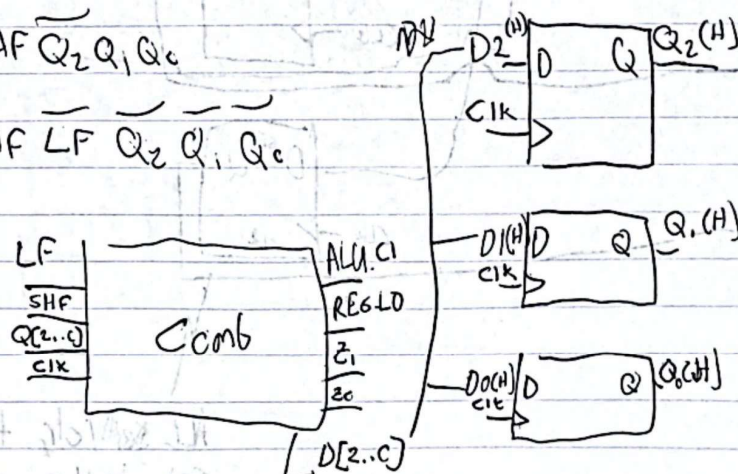
$$DO \leftarrow \text{LF SHF} \overline{Q_2} \overline{Q_1} \overline{Q_0} + \text{SHF} \overline{Q_2} \overline{Q_1} \overline{Q_0} + \text{LF} \overline{Q_2} \overline{Q_1} \overline{Q_0}$$

$$\text{ALU.CI} \leftarrow \overline{Q_2} \overline{Q_1} \overline{Q_0} + \text{SHF} \overline{Q_2} \overline{Q_1} \overline{Q_0} + \text{SHF}$$

$$\text{REG.LD} \leftarrow \overline{Q_2} \overline{Q_1} \overline{Q_0}$$

$$Z_1 \leftarrow \text{SHF} \overline{Q_2} \overline{Q_1} \overline{Q_0}$$

$$Z_0 \leftarrow \text{SHF LF} \overline{Q_2} \overline{Q_1} \overline{Q_0}$$



Decoder Redo:

```
1  library ieee; use ieee.std_logic_1164.all;
2
3  -- This code is the redesign of the decoder from Lab 2, this was written by Evan Baesler
4  -- This code uses MSOP to create active-low LED outputs for a 7-segment display, from 0 to F in hex.
5
6  entity DECODER is port(
7      X: in std_logic_vector(3 downto 0);
8      T_L: in std_logic;
9
10     A_L, B_L, C_L, D_L, E_L, F_L, G_L: out std_logic;
11 );
12
13 end DECODER;
14
15 architecture behavior of DECODER is
16     signal m0, m1, m2, m3, m4, m5, m6, m7, m8, m9, mA, mB, mC, mD, mE, mF: std_logic;
17     signal A_H, B_H, C_H, D_H, E_H, F_H, G_H: std_logic;
18     signal T_H: std_logic;
19
20 begin
21
22     A_L <= not A_H;
23     B_L <= not B_H;
24     C_L <= not C_H;
25     D_L <= not D_H;
26     E_L <= not E_H;
27     F_L <= not F_H;
28     G_L <= not G_H;
29     T_H <= not T_L;
30
31     m0 <= not X(3) and not X(2) and not X(1) and not X(0);
32     m1 <= not X(3) and not X(2) and not X(1) and X(0);
33     m2 <= not X(3) and not X(2) and X(1) and not X(0);
34     m3 <= not X(3) and not X(2) and X(1) and X(0);
35     m4 <= not X(3) and X(2) and not X(1) and not X(0);
36     m5 <= not X(3) and X(2) and not X(1) and X(0);
37     m6 <= not X(3) and X(2) and X(1) and not X(0);
38     m7 <= not X(3) and X(2) and X(1) and X(0);
39     m8 <= X(3) and not X(2) and not X(1) and not X(0);
40     m9 <= X(3) and not X(2) and not X(1) and X(0);
41     mA <= X(3) and not X(2) and X(1) and not X(0);
42     mB <= X(3) and not X(2) and X(1) and X(0);
43     mC <= X(3) and X(2) and not X(1) and not X(0);
44     mD <= X(3) and X(2) and not X(1) and X(0);
45     mE <= X(3) and X(2) and X(1) and not X(0);
46     mF <= X(3) and X(2) and X(1) and X(0);
47
48
49     A_H <= T_H or (m0 or m2 or m3 or m5 or m6 or m7 or m8 or m9 or mA or mC or mE or mF);
50     B_H <= T_H or (m0 or m1 or m2 or m3 or m4 or m7 or m8 or m9 or mA or mD);
51     C_H <= T_H or (m0 or m1 or m3 or m4 or m5 or m6 or m7 or m8 or m9 or mA or mB or mD);
52     D_H <= T_H or (m0 or m2 or m3 or m5 or m6 or m8 or m9 or mB or mC or mD or mE);
53     E_H <= T_H or (m0 or m2 or m6 or m8 or mA or mB or mC or mD or mE or mF);
54     F_H <= T_H or (m0 or m4 or m5 or m6 or m8 or m9 or mA or mB or mC or mE or mF);
55     G_H <= T_H or (m2 or m3 or m4 or m5 or m6 or m8 or m9 or mA or mB or mD or mE or mF);
56 end behavior;
```

Figure 1: VHDL for new Quartus implementation.

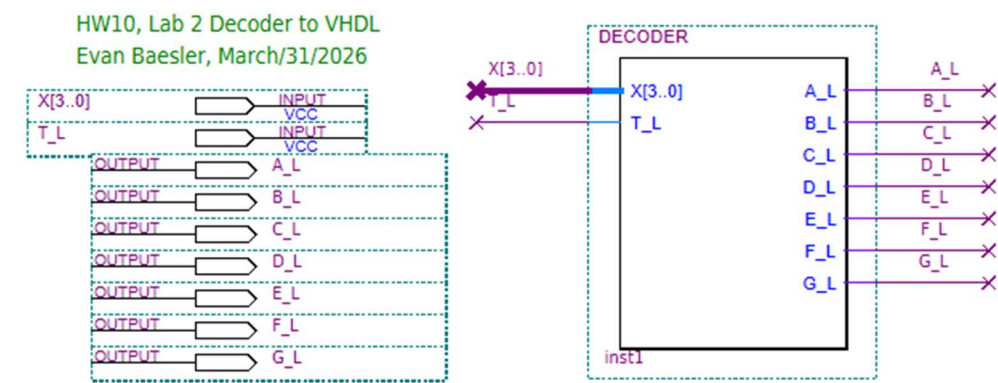


Figure 2: Quartus BDF for VHDL Decoder implementation.

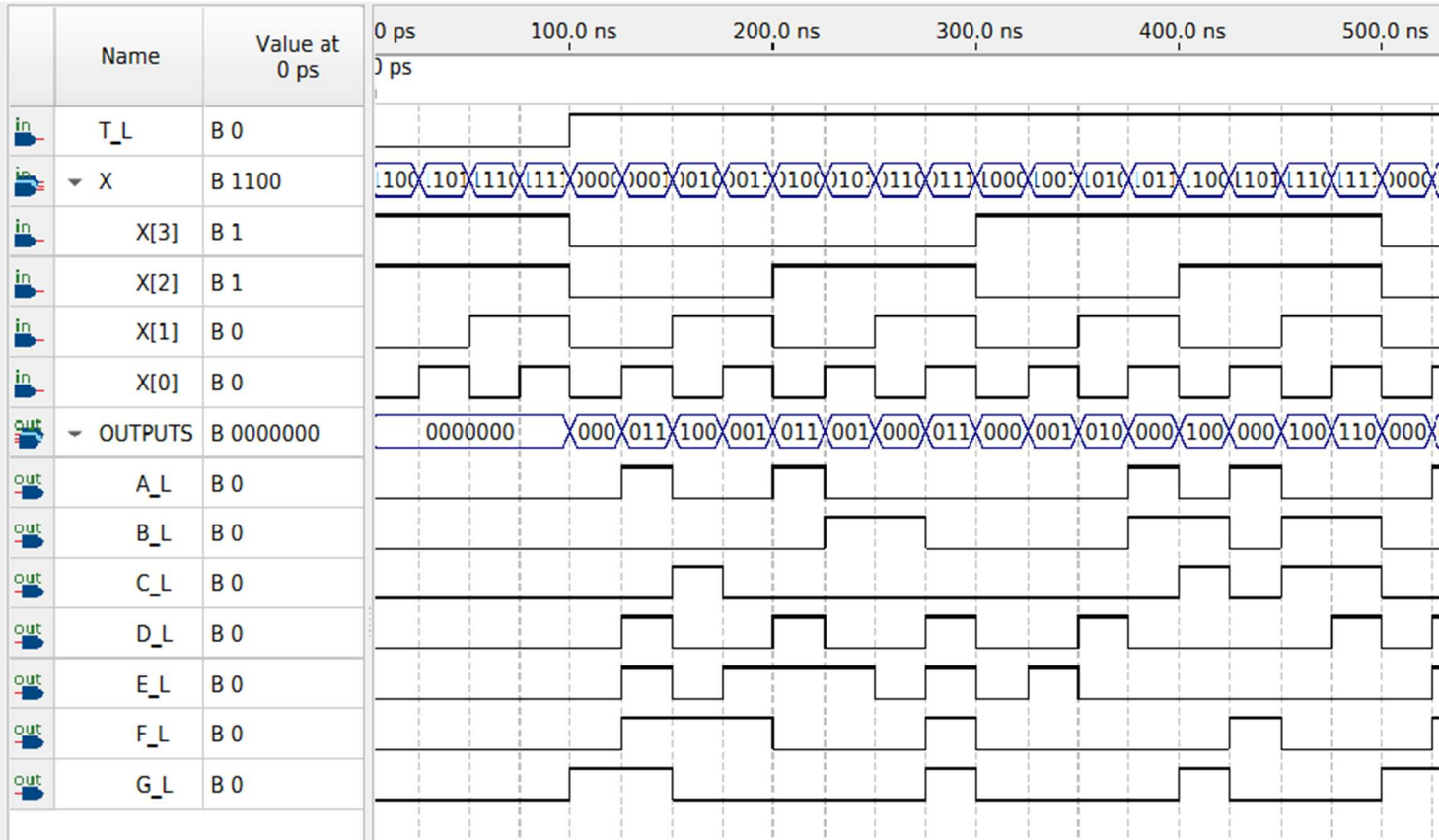


Figure 3: Quartus Simulation for VHDL Decoder implementation, matches annotated Lab 2 diagram.